

Program 1: Maven Project Initialization:

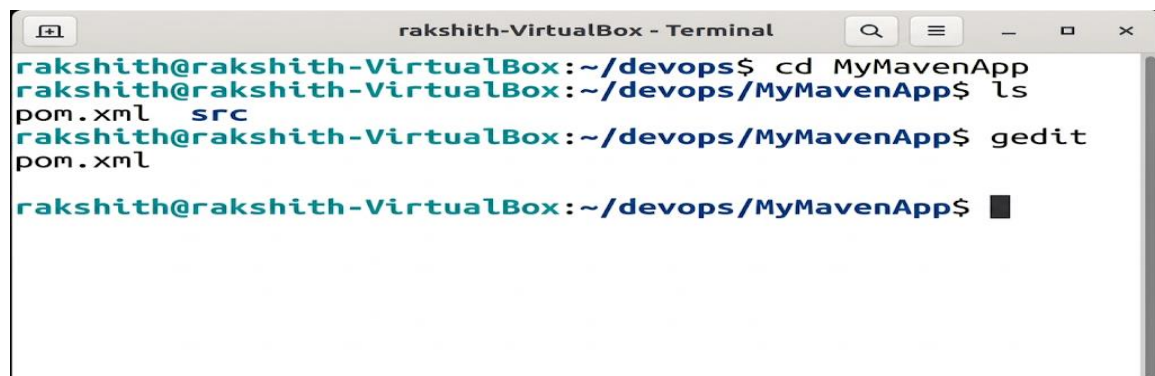
Step 1: Create a Maven project:

`mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false`

```
rakshith@rakshith-VirtualBox:~$ cd devops
rakshith@rakshith-VirtualBox:~/devops$ mvn archetype:generate -groupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 6.9 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/rakshith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/rakshith/devops/MyMavenApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 5.299 s
[INFO] Finished at: 2026-03-19T13:15:32+05:30
[INFO]
```

This command generates a basic Maven project structure with a sample `App.java` and `AppTest.java`.

Step 2: Go to the project directory using command `cd MyMavenApp`:



```
rakshith@rakshith-VirtualBox:~/devops$ cd MyMavenApp
rakshith@rakshith-VirtualBox:~/devops/MyMavenApp$ ls
pom.xml  src
rakshith@rakshith-VirtualBox:~/devops/MyMavenApp$ gedit pom.xml
rakshith@rakshith-VirtualBox:~/devops/MyMavenApp$
```

Step 3: Edit `pom.xml`: `gedit pom.xml`

- Go to mvnrepository.com
- Search “junit”
- Copy dependency
- Paste inside



```
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3   <modelVersion>4.0.0</modelVersion>
4   <groupId>com.example</groupId>
5   <artifactId>MyMavenApp</artifactId>
6   <packaging>jar</packaging>
7   <version>1.0-SNAPSHOT</version>
8   <name>MyMavenApp</name>
9   <url>http://maven.apache.org</url>
10  <!-- Properties: Customize Java version or plugin versions -->
11  <properties>
12    <maven.compiler.source>11</maven.compiler.source>
13    <maven.compiler.target>11</maven.compiler.target>
14  </properties>
15  <dependencies>
16    <dependency>
17      <groupId>junit</groupId>
18      <artifactId>junit</artifactId>
19      <version>3.8.1</version>
20      <scope>test</scope>
21    </dependency>
22  </dependencies>
23  <!-- Build: Configuring plugins and build settings -->
24  <build>
25    <plugins>
26      <!-- Example: Maven Compiler Plugin to compile Java code -->
27      <plugin>
28        <groupId>org.apache.maven.plugins</groupId>
29        <artifactId>maven-compiler-plugin</artifactId>
30        <version>3.8.1</version>
31        <configuration>
32          <source>1.7</source>
33          <target>1.7</target>
34        </configuration>
35      </plugin>
36    </plugins>
37  </build>
38 </project>
```

Update App.java:

Step 4: Build the project using command mvn clean install The project building will start like this.

```

rakshith@rakshith-VirtualBox:~/devops/MyMavenApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/rakshith/devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith/devops/MyMavenApp/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/rakshith/devops/MyMavenApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith/devops/MyMavenApp/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.028 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---
[INFO] Building jar: /home/rakshith/devops/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenApp ---
[INFO] Installing /home/rakshith/devops/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/rakshith/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/rakshith/devops/MyMavenApp/pom.xml to /home/rakshith/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.pom
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.495 s
[INFO] Finished at: 2026-03-19T13:17:37+05:30
[INFO]

```

The “Build Success” message ensures that our Maven project is built successfully.

Step 7: To run the Maven application

- Execute the JAR file with the command:

```
java -jar MyMavenApp-1.0-SNAPSHOT.jar
```